



Movement Metrics



Smithsonian Conservation
Biology Institute

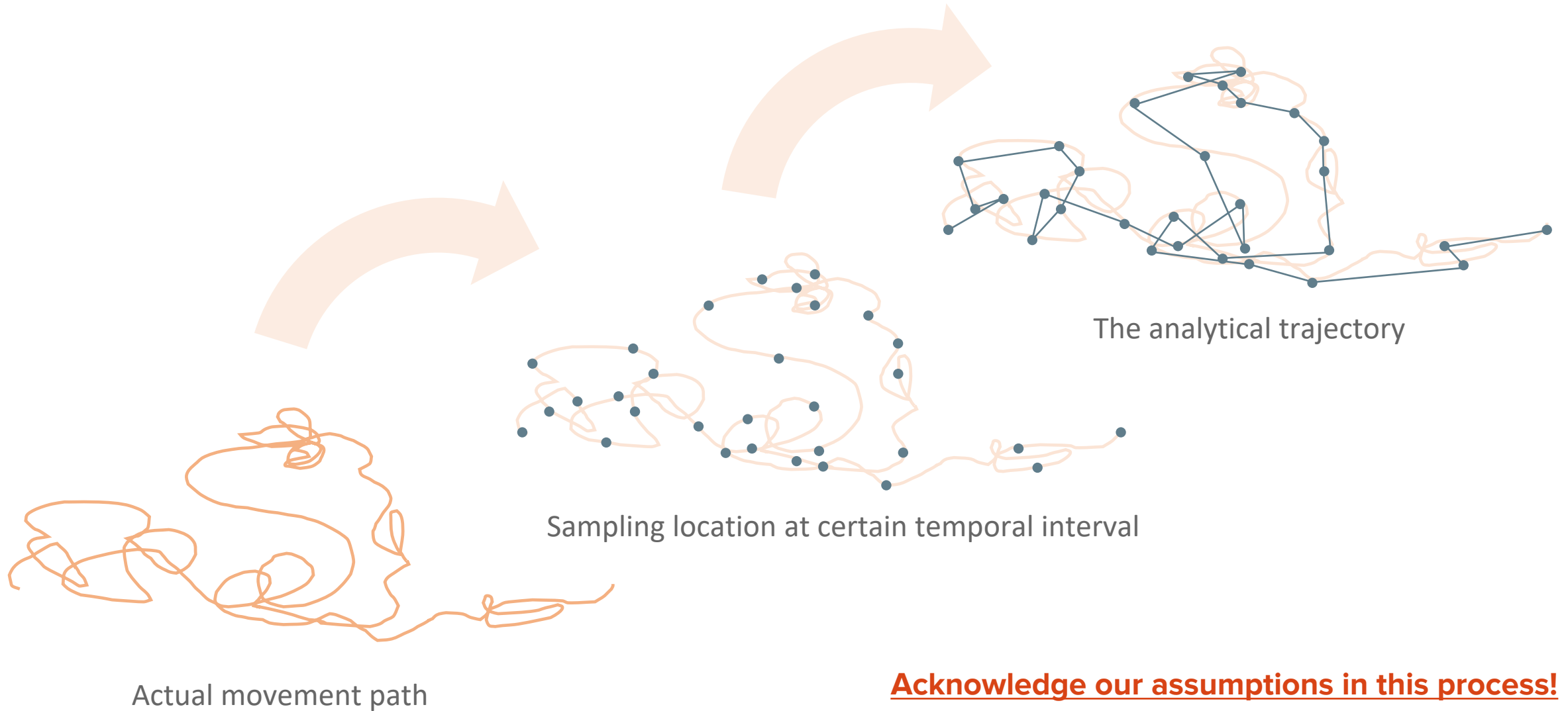


SENCKENBERG
NATURFORSCHUNG



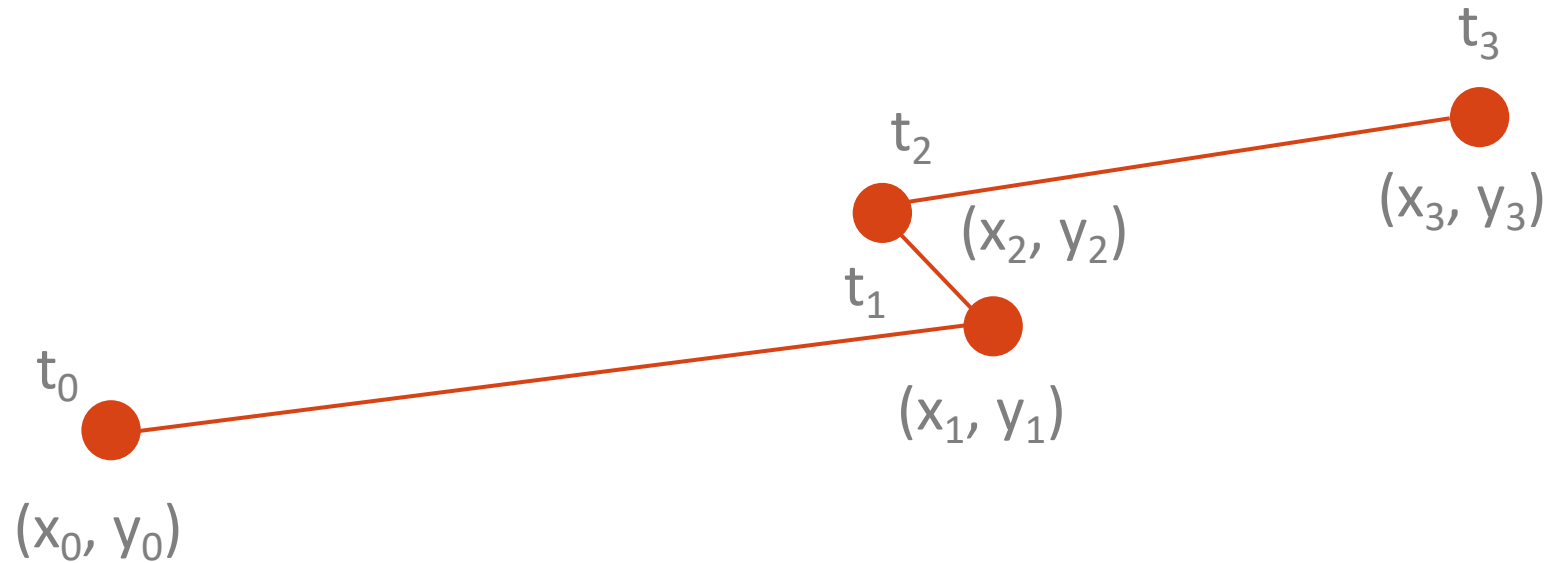
Jared Stabach and Wenjing Xu
@ Fudan University | August 4 - 10, 2025

From raw data to analytical track



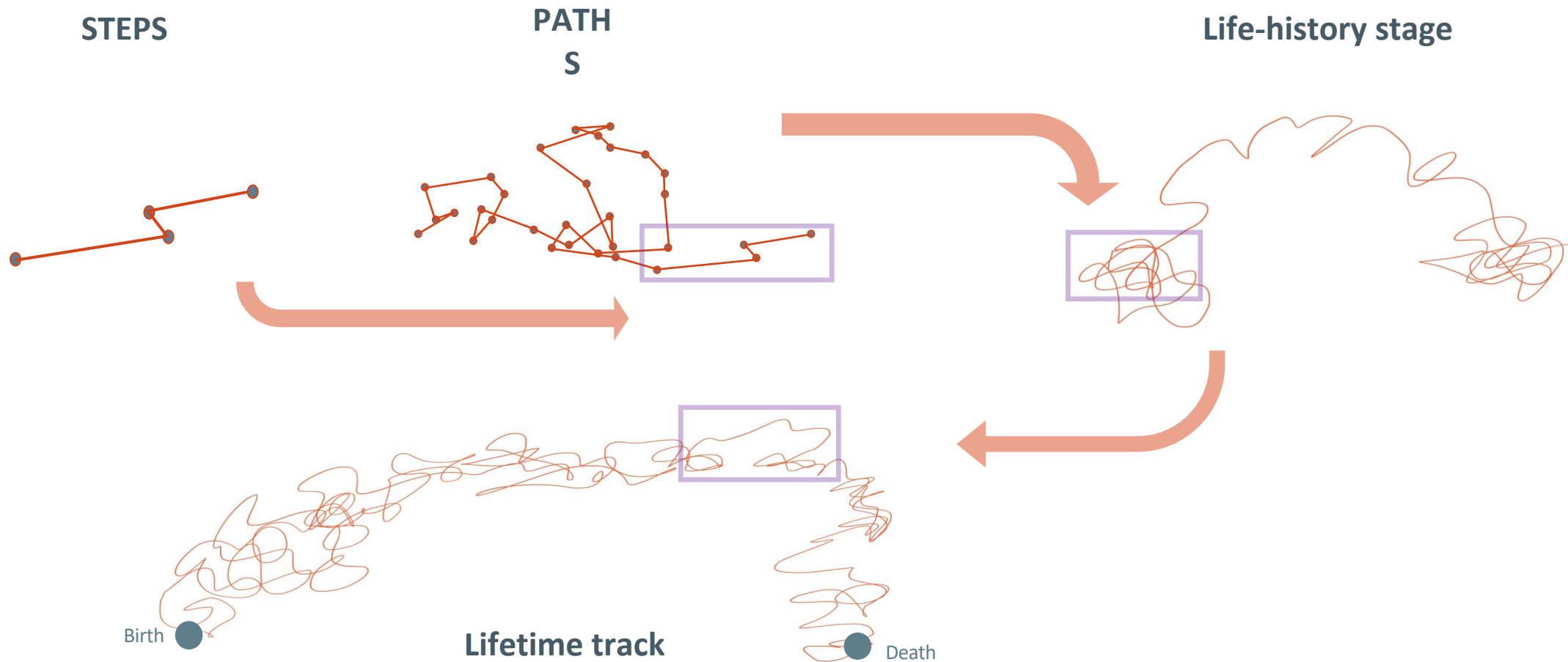
From raw data to analytical track

Movement is locations through time



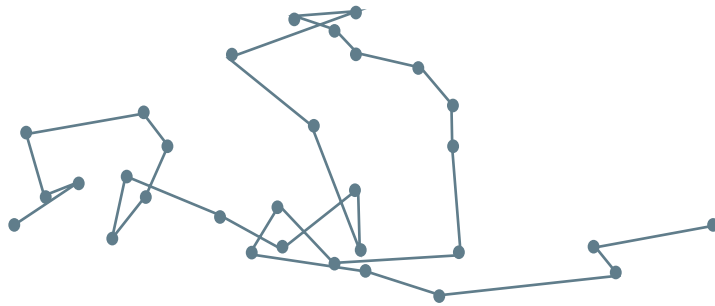
From raw data to analytical track

The ecological meaning is highly scale-dependent



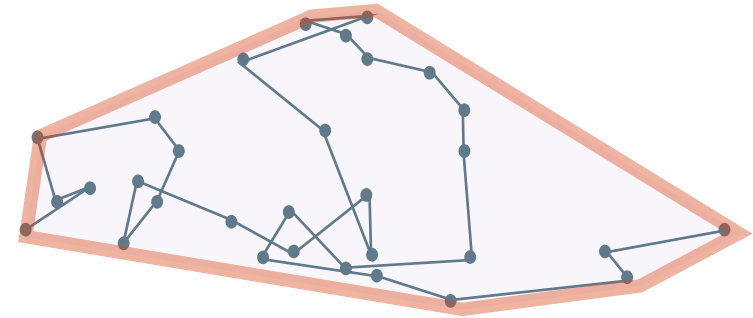
Movement metrics as the first analytical step

TRAJECTORY/PATH-LEVEL METRICS



- Step length
- Turn angle
- Displacement
- First passage time
- Time to return
- Straightness
- ...
- Turn angle correlation
- Net-squared displacement

SPACE-USE METRICS



- Range estimate
 - MCP
 - KDE
 - LoCoH
 - dBBMM
 - ...
- Area use intensity
- Home range fidelity
- ...

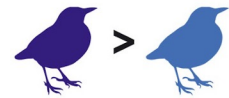
Always think about scale - does your data match your question?

The frequency, gaps, and shifts of movement data affects the calculation and comparability of movement metrics.

Higher resolution

(5 s intervals)

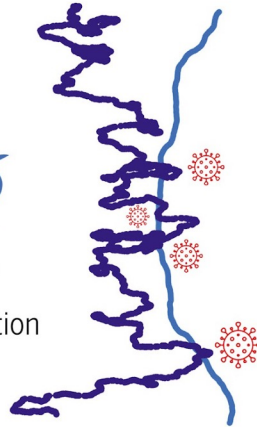
Exploration



Bold

Shy

Multiple interaction hotspots



Individual variation:

Are shy birds less explorative than bold ones?

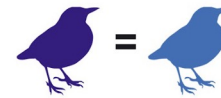
Biotic interactions:

What is the potential for disease transmission?

Lower resolution

(30 min intervals)

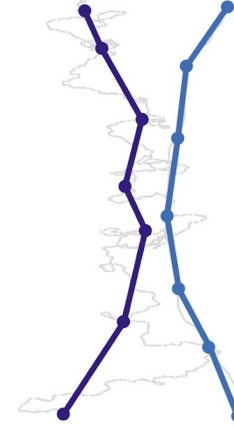
Exploration



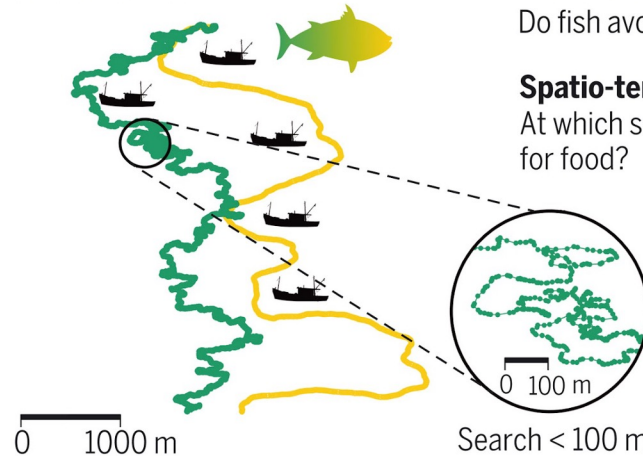
Bold

Shy

No interactions



Fish **avoid** vessels



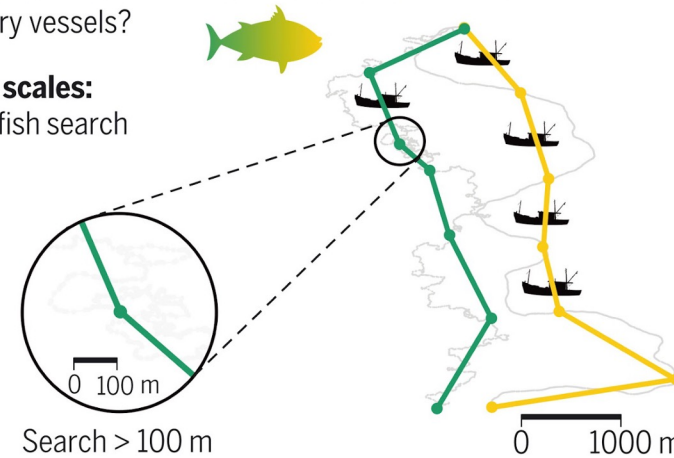
Interactions with humans:

Do fish avoid fishery vessels?

Spatio-temporal scales:

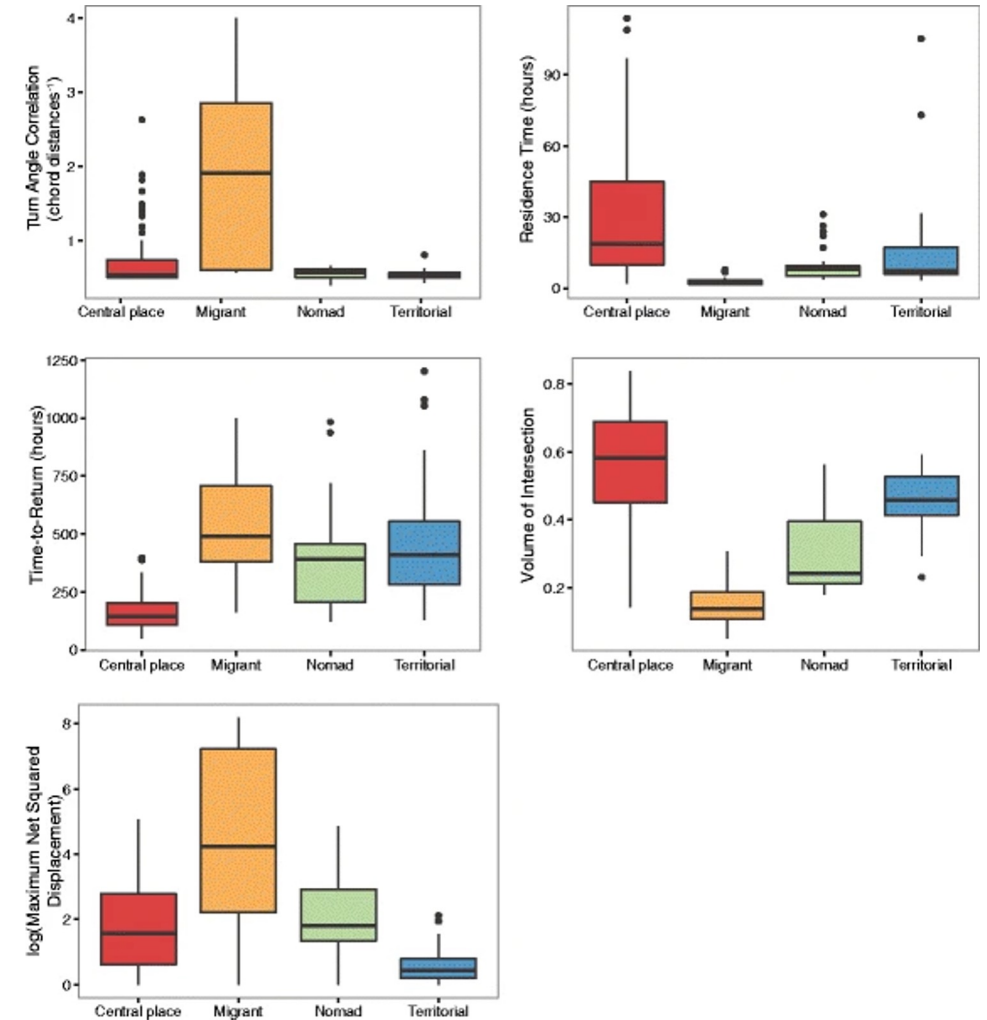
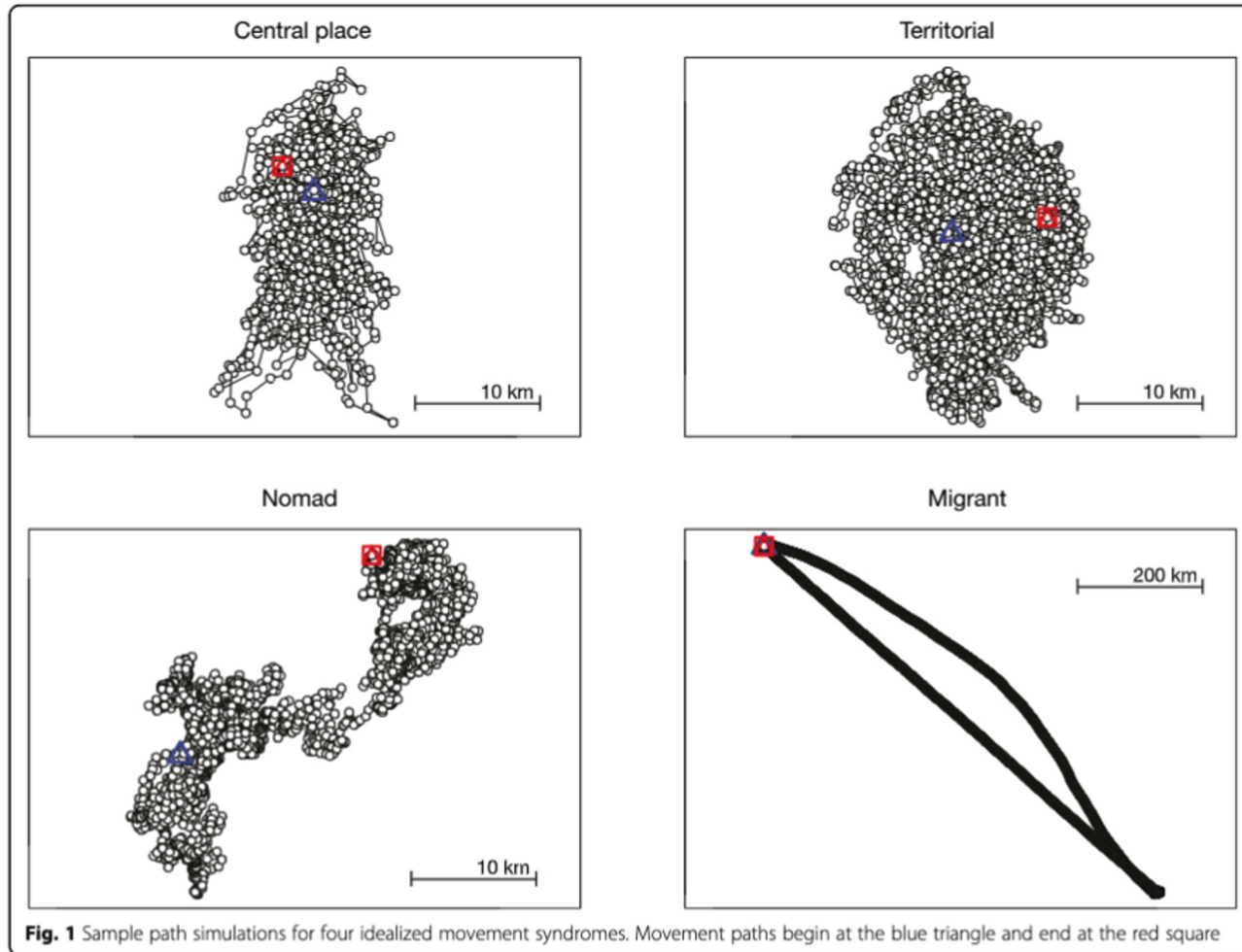
At which scale do fish search for food?

Fish do **not avoid** vessels



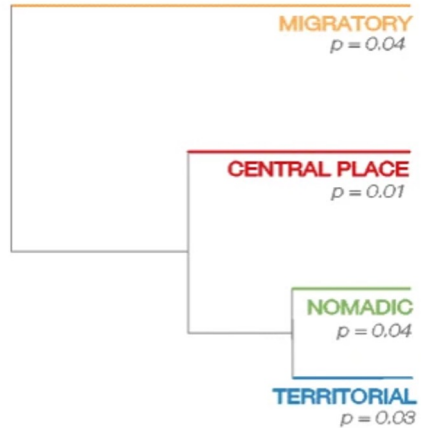
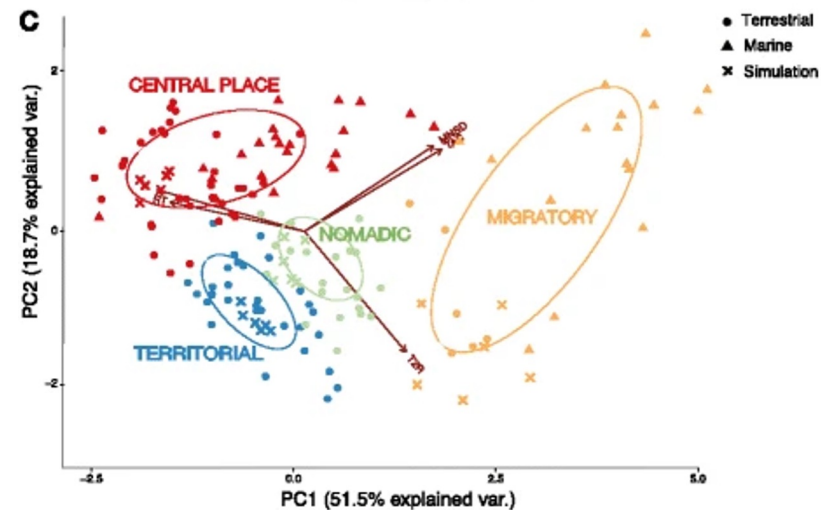
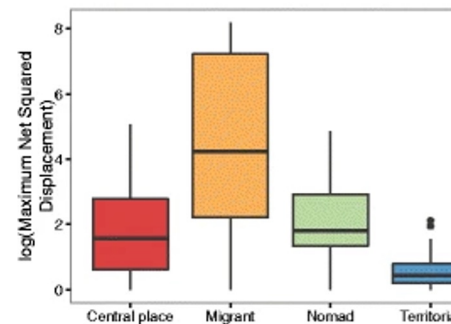
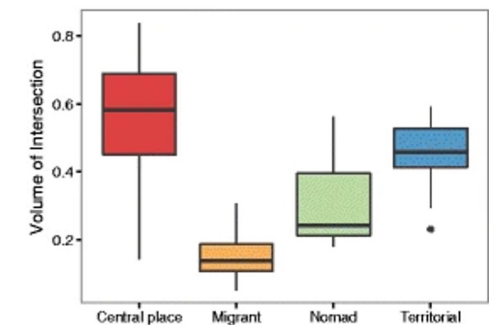
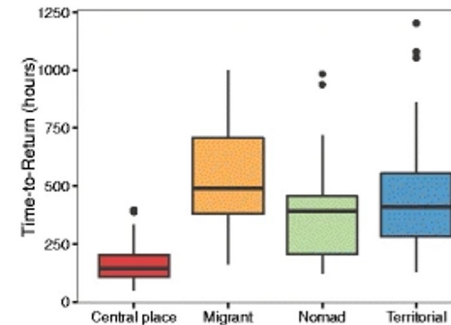
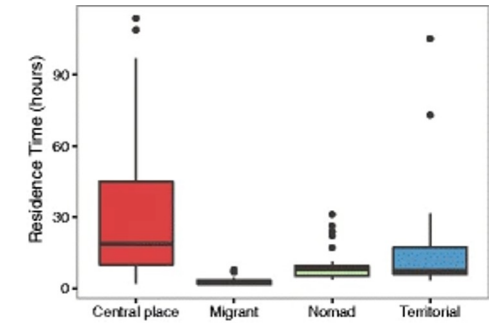
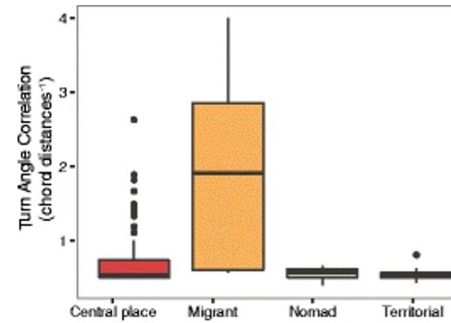
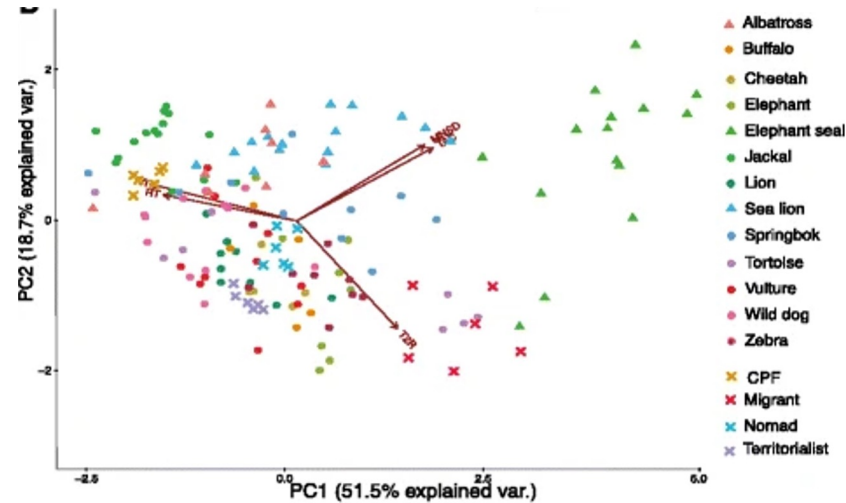
Movement metrics underlie behaviors (case 1)

clustering of movement metrics



Movement metrics underlie behaviors (case 1)

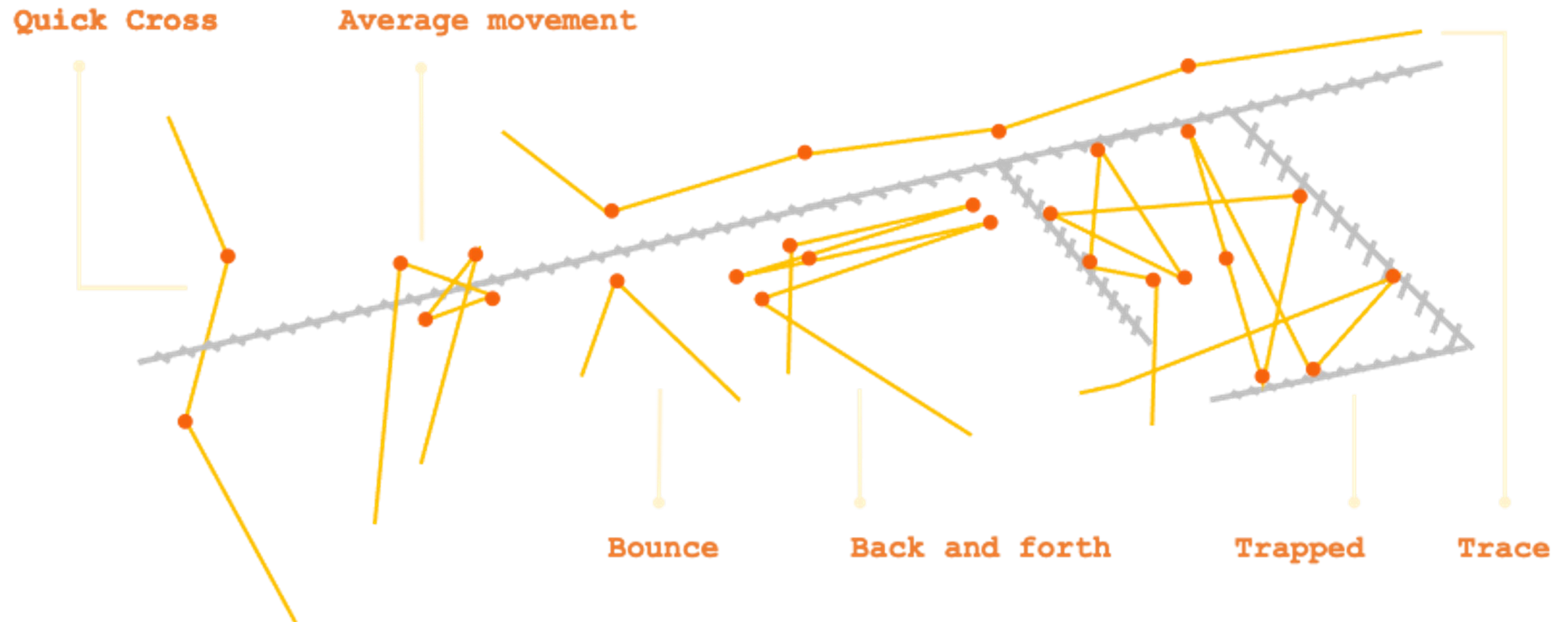
clustering of movement metrics





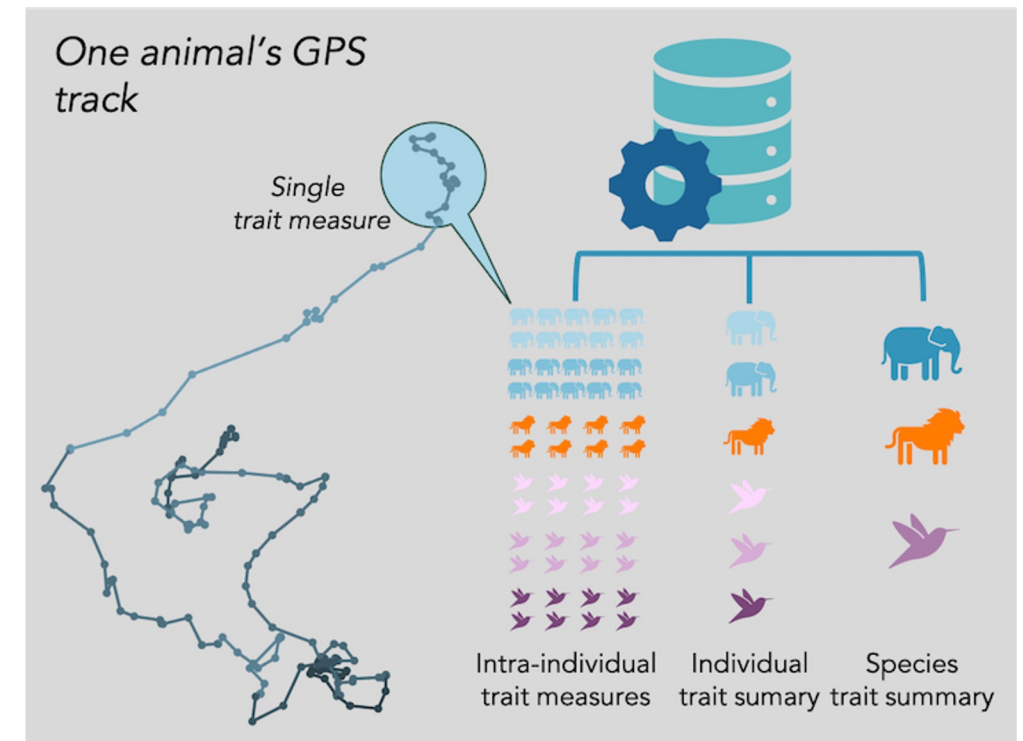
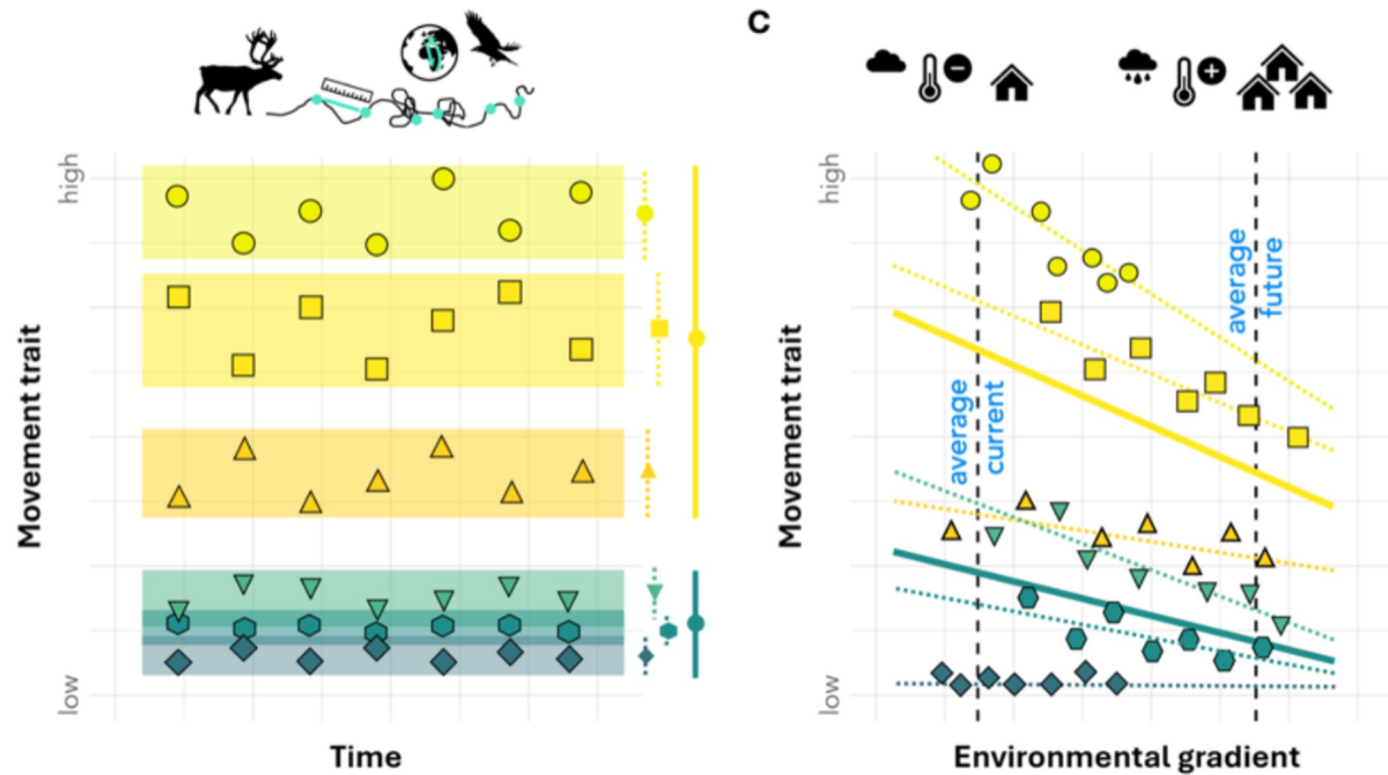
Movement metrics underlie behaviors (case 2)

Barrier Behavior Analysis



From movement metrics to movement traits

Standardized and comparable movement metrics to facilitate trait-based research on species interactions, community structure, and ecosystem functioning, and to support conservation and policy goals.



Questions?